

**Fifth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

CONTROL SYSTEMS

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
 - (2) All questions carry marks as indicated against them.
 - (3) Use graph paper wherever necessary.
 - (4) Assume suitable data and illustrate answers with neat sketches wherever necessary.
1. (a) Using the block reduction technique, determine the function C/R of a system represented by the block in Fig. 1

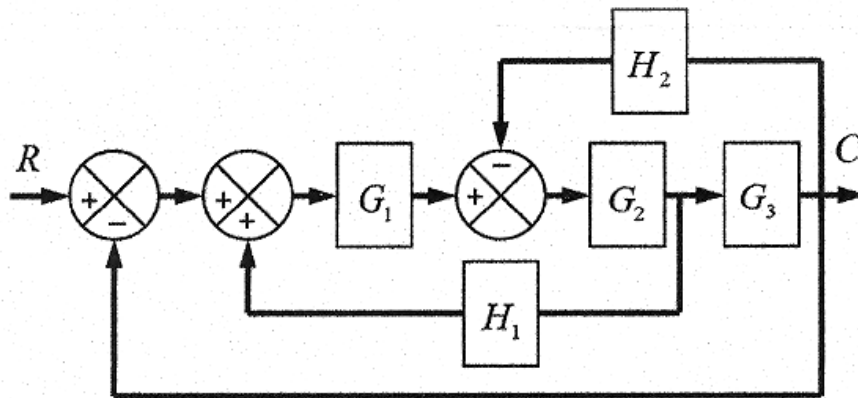


Fig. 1

6(CO2)

- (b) For the control system whose signal flow graph is shown in Fig. 2, using Mason's gain formula, find the system transfer function $Y(s)/R(s)$.

[Fig On Next page]

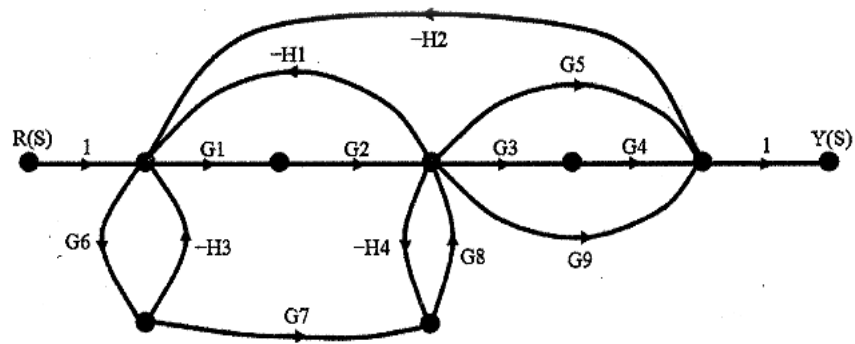


Fig. 2

6(CO2)

2. (a) The forward path gain of a unity feedback system is,

$$G(s) = \frac{81}{s(s + 7.2)}$$

Determine the unit step response of the system.

6(CO4)

- (b) For a second order control system, the unit step response is found to have a maximum overshoot of 18% and peak time of 0.25 second. Find the natural and damped frequency of oscillations and the location of second order poles.

6(CO4)

3. (a) Comment on the stability of a control system using Routh's criterion having following characteristics equation :

$$s^6 + s^5 + 5s^4 + 3s^3 + 2s^2 - 4s - 8 = 0$$

6(CO3)

- (b) Construct the root locus for a system with open-loop transfer function.

$$G(s)H(s) = \frac{K}{s(s + 1)(s + 3)}$$

6(CO3)

4. (a) Sketch the Bode plot for the open-loop transfer function of a unity feedback system given below and assess stability

$$G(s) = \frac{50}{(s + 1)(s + 2)}$$

6(CO4)

- (b) What is polar plot ? What are the steps to draw a polar plot ? Enlist the advantages, limitations and applications of polar plot. 6(CO3)

5. (a) Develop the electrical network shown in Fig. 3 below by a state model

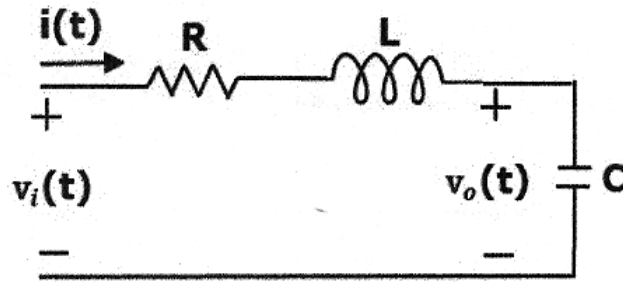


Fig. 3

6(CO5)

- (b) Compare linear and non-linear control systems.

6(CO1)

